CIRCULAR MOTION.

Motion of a body on a circular path is called circular motion.

EXAMPLES.

(i) Motion of planets around the sun.
(ii) A stone whirled around by a string.
(iii) A car turning around a corner.
(iv) Satellites in orbits around the earth.

During circular motion, velocity of body changes continuously. This change in velocity per unit time gives acceleration which is produced by diggerent forces.

In motion of planet around the sun, this acceleration is produced by gravitational force. The coulomb force of attraction is responsible for ... of electron around the nucleus.

ANGULAR DISPLACEMENT.

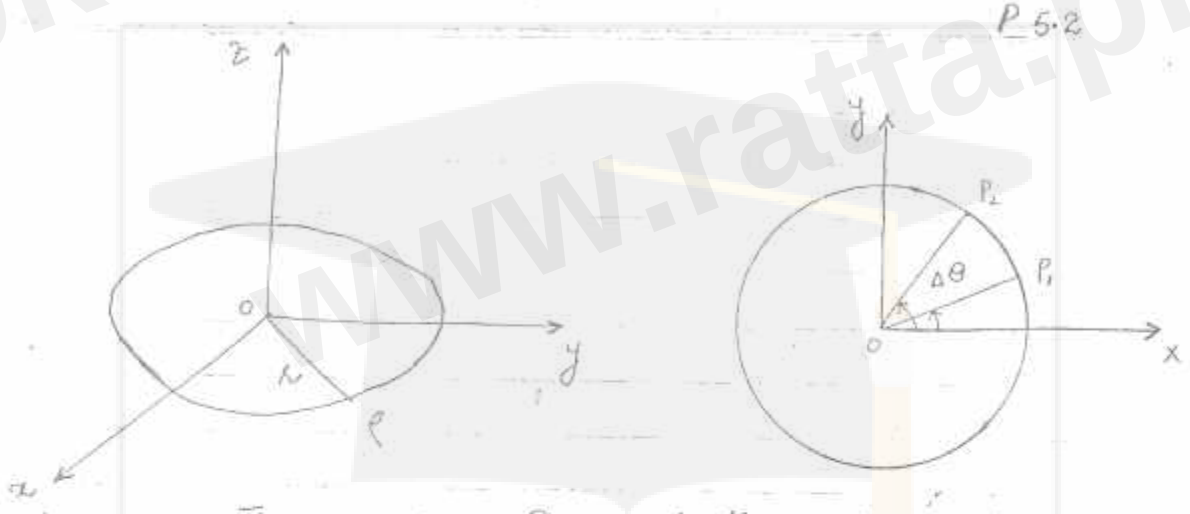
Angle ($\Delta\theta$) swept by a body in time interval Δt during the angular motion, at centre of its circular path is called "Angular Displacement"

EXPLANATION.

Consider the motion of a particle 'P' attached at the end of mauless rigid rod of length 'r' whose other end is pivoted at the centre 'O' of the circular path. Suppose the particle 'P' moves in a circle of radius 'r' whose centre lies at the origin of xy-plane.

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P_5.2



The axis of rotation is along Z-axis with pivat 'O' as the origin of co-ordinates. Let,

$\vec{OP_1}$ = Position vector of particle at time 't' making angle θ with x-axis.

$\vec{OP_2}$ = Position vector of particle at time $(t + \Delta t)$ making angl $(\theta + \Delta\theta)$ with X-axis.

From figure,

$P_1 \hat{O} P_2 = \Delta\theta$ = Angular disp of particle 'P' during time interval "Δt"

SIGN OF CONVENTION.

(i) The angular displacement $\Delta\theta$ is taken as positive, when the rotation of particle 'P' is "Anti-clockwise"

(ii) The angular displacement $\Delta\theta$ is taken as negative, when the rotation of particle 'P' is taken as "Clockwise".

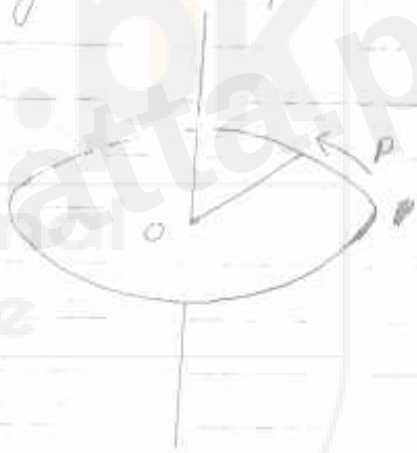
small value of $\Delta\theta$, the angular disp. is a vector quantity.

The direction associated with $\Delta\theta$ is along the axis of rotation and is given by "Right Hand Rule"

RIGHT HAND RULE:

"According to this rule, "Grasp the axis of rotation in right hand such that curl of fingers is in the direction of rotation. Then thumb points in the direction of angular displacement."

According to right hand rule the direction of angular disp. is out of the plane of circle, when the particle rotated in anti-clockwise direction.



UNITS: The units of angular displacement are

(i) Degree
(ii) Revolution
(iii) Radian

RADIAN:

It is SI-unit of angular disp. It is defined as, "Angle made by an arc whose length is equal to the radius of circle is called radian"

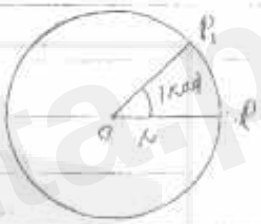
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From Figure,

when arc $PP_1 = r$

and $P\hat{O}P_1 = 1$ radian



PROVE THAT $S = r\theta$.

Here,

S = Arc Length

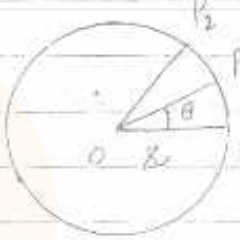
r = Radius of circular path.

θ = Angle made by an arc in radians at the centre of circular path.

PROOF:

Consider the motion of the particle 'P' on a circular path of radius 'r'.

For motion from P to P_1, the arc $PP_1 = S$ makes angle 'θ' measured in radians at the centre of the circle.



Also,

Length of arc $PP_1 = r$

So that, $P\hat{O}P_1 = 1$ radian

Diagrame shows that angle subtended at the centre of the circle is proportional to length of arc

So,

$$\frac{\text{arc } PP_1}{\text{arc } PP_2} = \frac{\theta \text{ radian}}{1 \text{ radian}}$$

$$\Rightarrow \quad \frac{S}{r} = \frac{\theta}{1}$$

$$\Rightarrow \quad \boxed{S = r\theta}$$

which is required relation.

SHOW THAT 1 radian = 57.3°:

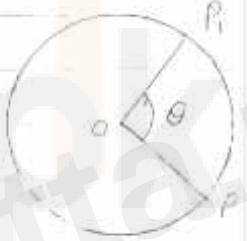
PROOF:

As we know that

$$S = r\theta$$

$$\Rightarrow \quad \theta = \frac{S}{r} \text{ radian}$$

Consider a particle moves in a circle of radius "r" and covers distance $S = 2\pi r$ in its one revolution.



$$\therefore \quad \theta = \frac{2\pi r}{r} \text{ radian}$$

Here,

1 revolution = 2π radians

Also,

1 revolution = 360 Degrees

$$\therefore \quad 2\pi \text{ radians} = 360 \text{ Degrees}$$

$$1 \text{ radian} = \frac{360^\circ}{2\pi}$$

$$\boxed{1 \text{ radian} = 57.3^\circ}$$

which is required relation.

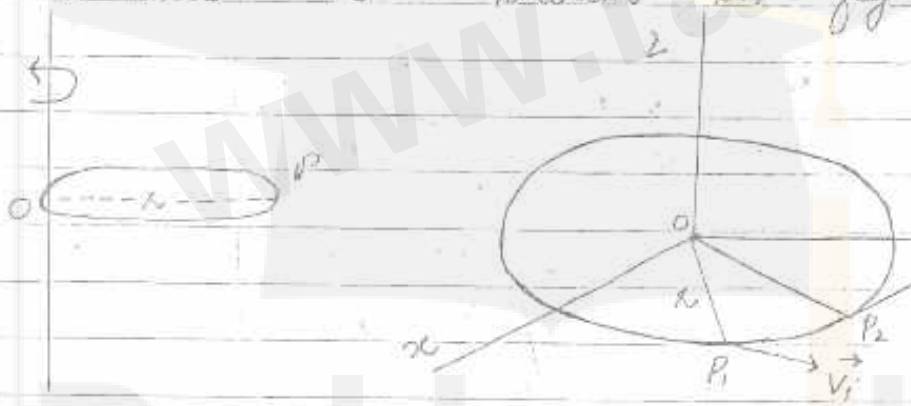
ANGULAR VELOCITY:

The rate of change of angular displacement is called angular

PROVE THAT $a = r\alpha$:

PROOF:

Consider a rigid body rotating about z-axis as shown in figure.



Let a point P in the rigid body rotates in circle of radius r whose centre lies at the origin of xy-plane.

Suppose the point P moves from point P_1 to P_2 in time interval Δt. $\vec{v}_i$ and $\vec{v}_f$ are instantaneous linear velocities of point P at instants "t_i" and "t_f" respectively.

Now,

$$v_i = r\omega_i \quad \text{(i)}$$

$$v_f = r\omega_f \quad \text{(ii)}$$

∴ We can write,

$$v_f - v_i = r(\omega_f - \omega_i)$$

Divide B.S by $t_f - t_i$

$$\frac{v_f - v_i}{t_f - t_i} = r\left(\frac{\omega_f - \omega_i}{t_f - t_i}\right)$$

OR

$$\frac{\Delta V}{\Delta t} = r\frac{\Delta \omega}{\Delta t}$$

Taking Limit $\Delta t \to 0$ B.S.

$$\lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = r \lim_{\Delta t} \frac{\Delta \omega}{\Delta t} \quad \text{---(iii)}$$

As,

$$\lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = a$$

and,

$$\lim_{\Delta t \to 0} \frac{\Delta \omega}{\Delta t} = \alpha$$

Put these values in eq. (2)

$$\therefore \boxed{a = r\alpha}$$

where,

a = Linear or tangential acceleration of point P of rigid body

In vector form above eq. can be written as,

$$\boxed{\vec{a} = \vec{\alpha} \times \vec{r}}$$

Comparison Of Linear And Angular Equations:

The equations of angular motion are exactly analogous to those in linear motion.

Linear Motion	Angular Motion
(i) $v_f = v_i + at$	(i) $\omega_f = \omega_i + \alpha t$
(ii) $2aS = v_f^2 - v_i^2$	(ii) $2\alpha\theta = \omega_f^2 - \omega_i^2$
(iii) $S = v_i t + \frac{1}{2} a t^2$	(iii) $\theta = \omega_i t + \frac{1}{2} \alpha t^2$
(iv) $v_{av} = \frac{v_i + v_f}{2}$	(iv) $\langle \omega \rangle = \frac{\omega_i + \omega_f}{2}$
(v) $F = ma$	(v) $\tau = I\alpha$

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CENTRIPETAL FORCE:

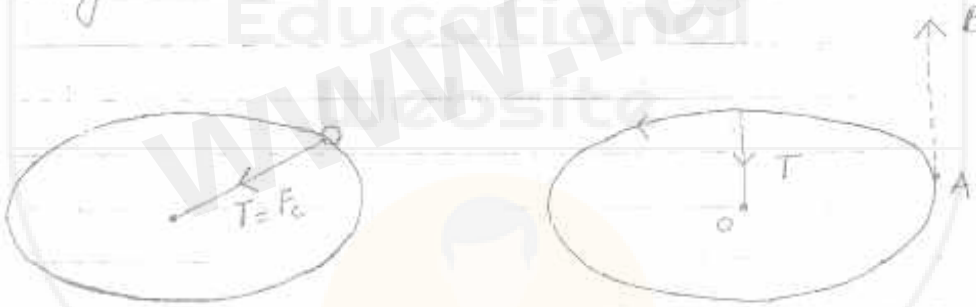
The force required to keep a body revolving on a circular path and is always directed towards the centre of circle is called "Centripetal Force"

OR "

The force required to bend the normally straight path of the particle into a circular path is called the Centripetal force."

EXAMPLE:

When a ball is whirled in a horizontal circle at the end of a string. Then tension in a string believes as centripetal force in order to keep the ball moving in circle.

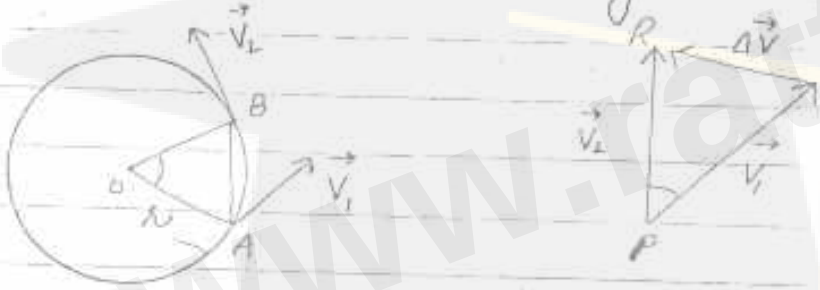


If the string breaks, when the ball is at point A, the ball will follow straight line path AB. This shows that ball will not continue along the circular path in the absence of tension force which acts as centripetal force.

DETERMINATION:

Consider a particl of mass 'm' moving on a circular path

of radius 'r' with uniform linear speed 'V' and constant angular velocity 'ω'.



As the particle moves from A to B with uniform speed 'V' but the velocity of the particle changes because of change in direction. Hence, the acceleration of the particle is,

$$a = \frac{\Delta V}{\Delta t} \longrightarrow (i)$$

Where 'Δt' is time taken by the particle to travel from A to B, all $\vec{V}_1$ and $\vec{V}_2$. Now time taken to travel distance is,

$$\Delta t = \frac{S}{V} \longrightarrow (ii)$$

By putting value of 'Δt' in (i)

$$\therefore \quad a = \frac{\Delta V}{S/V}$$

$$\Rightarrow \quad a = V\,\frac{\Delta V}{S} \longrightarrow (iii)$$

Let us draw a triangle PQR as shown in fig.

$$\overrightarrow{QR} = \Delta\vec{V} = \text{Change in vel.} = \vec{V}_2 - \vec{V}_1$$

Now,

$$m\angle AOB = m\angle QPR \quad \because [\text{Angle b/w lines}]$$

As,

$$V_1 = V_2 \quad \text{and} \quad \bar{OA} = \bar{OB}$$

ΔOAB and ΔPQR are isosceles triangles and they are also similar. Hence we can write

$$\frac{\bar{AB}}{\bar{QR}} = \frac{\bar{OA}}{\bar{PQ}}$$

$$\Rightarrow \frac{\bar{AB}}{\Delta V} = \frac{r}{V} \longrightarrow (iv)$$

When $\Delta t \to 0$, the point 'B' is very close to point 'A' on circle. Then,

arc $\bar{AB} \simeq$ Line AB

So we can substitute $\bar{AB} = S$ in Eq. (iv)

$$\frac{S}{\Delta V} = \frac{r}{V}$$

or $$\Delta V = \frac{S\,V}{r}$$

By putting the value of ΔV in Eq. (iii)

$$a = \frac{V\,(S\,V/r)}{S}$$

$$\boxed{a = \frac{V^2}{r}} \quad (v)$$

This is the instantaneous acceleration produced by centripetal force and is called "centripetal acceleration".

It is denoted by a_c. This acceleration is directed along the radius towards the centre of the circle.

Multiply Eq. (v) B.S by 'm' we get,

$$ma = \frac{mv^2}{r}$$

According to 2nd law of motion,

$$F = ma$$

So,

$$\boxed{F_c = \frac{mv^2}{r}} \quad \text{(vi)}$$

which is required expression for centripetal force. As,

$$v = r\omega$$

$$\therefore F_c = \frac{m(r\omega)^2}{r}$$

$$\boxed{F_c = mr\omega^2}$$

This is expression of centripetal force in terms of angular velocity

EXAMPLES:

(i) Earth revolves around the sun because of the centripetal force provided by gravitational force between sun and earth.

(ii) When a car takes turn around a corner then centripetal force is provided by friction between road and tyres of car.

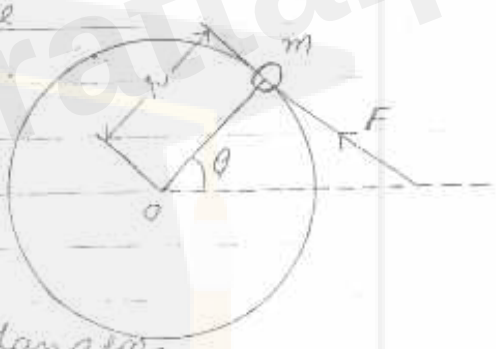
MOMENT OF INERTIA:

The product of mass and square of the moment arm is called "Moment of Inertia"

EXPLANATION:

Consider a mass 'm' attached to a massless rod pivoted at

'O'. Let us assume that pivot point 'O' is frictionless and system is in horizental plane. A force 'F' acts on the mass perpendicular to the rod and produce acceleration in the mass.



According to Newton's 2nd Law,

$$F = ma \longrightarrow (i)$$

The linear or tangential acceleration of mass 'm' is given by,

$$a = r\alpha$$

So, Eq. (i) becomes as,

$$F = mr\alpha \longrightarrow (ii)$$

Multiplying R.S by 'r'

$$Fr = mr^2\alpha$$

$$\tau = mr^2\alpha \quad \because (\tau = rF)$$

The quantity (mr^2) is known as moment of inertia represented by 'I'.

$$I = mr^2$$

The moment of inertia plays the same role in angular motion as the inertial mass in linear motion.

DEPENDANCE:

(i) Mass of a particle 'm'.

(ii) Distance of mass from axis of rotation.

UNITS:

SI-units of moment of inertia is $Kg.m^2$.

DIMENSION

$[ML^2]$ or $[ML^2T^0]$

MOMENT OF INERTIA OF RIGID BODY:

Consider a rigid body which is made up of n small pieces of masses i-e; $m_1, m_2 \ldots\ldots m_n$ at distances $r_1, r_2 \ldots\ldots r_n$ from axis of rotation 'O'.



Let the body is rotating with angular acceleration α. So the magnitude of torque acting on mass m_1 is,

$$T_1 = m_1 r_1^2 \alpha_1$$

Similarly, the torque on m_2 is,

$$T_2 = m_2 r_2^2 \alpha_2$$

And so on.

Since the body is rigid, so all the masses are rotating with the same angular acceleration α.

∴ Total Torque is given by,

$$T_{TOTAL} = (m_1 r_1^2 \alpha + m_2 r_2^2 \alpha + \ldots\ldots\ldots + m_n r_n^2 \alpha)$$

$$T_{TOTAL} = (m_1 r_1^2 + m_2 r_2^2 + \cdots\cdots + m_n r_n^2)\alpha$$

$$\left(\sum_{i=1}^{n} m_i r_i^2\right)\alpha$$

$$T = I\alpha$$

where,

I = Moment of inertia of rigid body

$$= \left(\sum_{i=1}^{n} m_i r_i^2\right)$$

ANGULAR MOMENTUM:

The magnitude of angular or rotational motion in a body is called its "Angular Momentum".

It is also defined as,

A particle is said to be possess an angular momentum about a reference axis if it moves such that its angular position changes relative to the reference axis.

Mathematically,

The cross product of position vector $\vec{r}$ and liner momentum $\vec{P}$ is called angular momentum.

$$\vec{L} = \vec{r} \times \vec{P}$$

Where $\vec{L}$ is angular momentum. It is a vector quantity.

EXPLANATION:

Consider a particle whose position is given by position vector $\vec{r}$

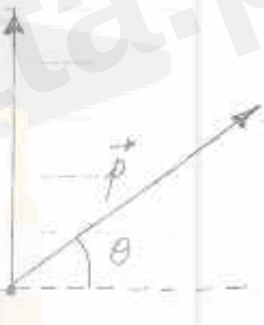
w.r.t an origin, $\dot{O}$.

Its linear momentum is,

$$\vec{P} = m\vec{v}$$

And, angular momentum is, $\vec{L} = \vec{r} \times \vec{P}$

Its magnitudes is, $|\vec{L}| = rp\sin\theta$



Where, θ is angle b/w $\vec{r}$ and $\vec{P}$.

The angular momentum $\vec{L}$ is perpendicular to the plane containing $\vec{r}$ and $\vec{P}$ and its direction is determined by right Hand rule as shown in fig.

As we know that,

$$\vec{L} = \vec{r} \times \vec{P}$$

$$\Rightarrow L = rp\sin\theta$$

$$= r(P\sin\theta)$$

$$\boxed{L = rP_L}$$



Where, $P_L = P\sin\theta$ is perpendicular concept component of $\vec{P}$ Lr to $\vec{r}$.

Also,

$$L = P(r\sin\theta)$$

$$\boxed{L = PL}$$



Where, $L = r\sin\theta$ is component of $\vec{r}$ Lr to $\vec{P}$.

Case (i) Eg $\theta = 0°$, $L = rp \sin 0° = 0$

Case (ii) Eg $\theta = 180°$, $L = rp \sin 180° = 0$

Case (iii) Eg $\theta = 90°$, $L = rp \sin 90° = rp$
$= rmv$

UNITS:

As,

$L = mvr$

$= Kg \cdot \frac{m}{sec} \cdot m$

$L = Kg\ m^2 / sec$

Or $L = Kg \frac{m}{sec^2} \cdot m \times m = N \cdot m \cdot sec^2$

$L = Joule\ sec^2$

DIMENSION:

As,

$L = mvr$

Dimension of 'L' is,

$[L] = \left[M \cdot \frac{L}{T} \cdot L\right]$

$[L] = [M L^2 T^{-1}]$

ANGULAR MOMENTUM AND MOMENT OF INERTIA:

As, the magnitude of angular momentum:

$L = mvr$

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NOTE:

We have seen that "$\vec{T}$" plays same role in rotatory motion as "$\vec{F}$" in linear motion. In linear motion if 'F' is the rate of change of linear momentum,

$$\vec{F} = \frac{\Delta \vec{P}}{\Delta t}$$

Then "$\vec{T}$" is the rate of change of angular momentum in rotatory motion.

$$\vec{T} = \frac{\Delta \vec{L}}{\Delta t}$$

LAW OF CONSERVATION OF ANGULAR MOMENTUM.

STATEMENT:

According to this law,

"If no external torque acts on a system, the total angular momentum of the system remains constant."

As we know that,

$$\vec{T} = \frac{\Delta \vec{L}}{\Delta t}$$

If no external torque acts on a system. Then,

$$\vec{T} = 0$$

$$\therefore \quad \frac{\Delta \vec{L}}{\Delta t} = 0$$

$$\Rightarrow \quad L = \text{constant}$$

As the angular momentum is constant so

$$I_1 \omega_1 = I_2 \omega_2$$

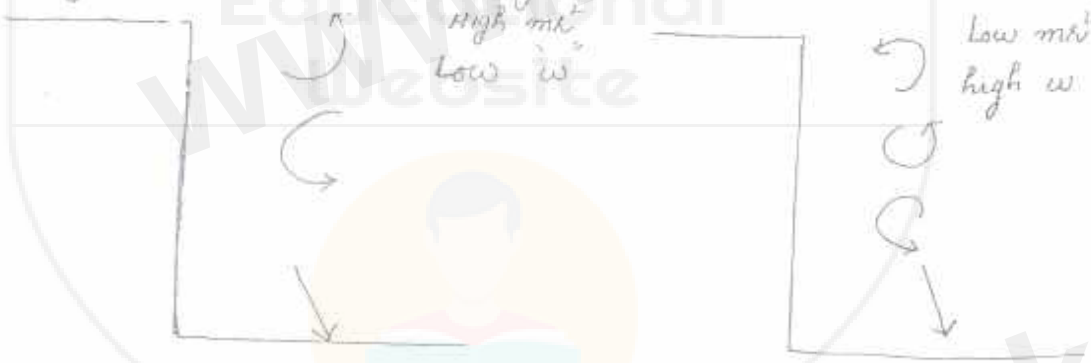
EXPLANATION:

For explanation, let us consider the following examples.

Examples:

(1) Let us take the example of a diver who jumps from a swimming board. If he curls his body, as the radius of his curled body decreases, the moment of inertia about his centre of gravity decreases and angular velocity increases to keep the angular momentum conserved.

In this way, he is able to perform certain number of before touching the water surface.



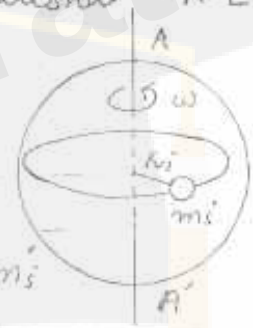
(2) The evolution of stars is considered to be the result of the law of conservation angular momentum.

ROTATIONAL KINETIC ENERGY:

The K.E possessed by a body due to its spin about its axis is known as "rotational kinetic energy"

EXPLANATION:

Consider a body is spinning about an axis with constant angular velocity 'ω', then each point of the body is moving in a circular path and has rotational K·E. Consider the spinning body is composed of tiny pieces of masses,



$m_1, m_2 \text{ ------- } m_n$

If a piece of mass m_i is at a distance r_i from axis of rotation. Then its K·E is,

$$K.E = \frac{1}{2} m_i V_i^2$$

where,

$$V_i = r_i \omega$$

$$\therefore K.E = \frac{1}{2} m_i (r_i \omega)^2$$

$$K.E = \frac{1}{2} m_i r_i^2 \omega^2$$

The rotational K·E of the spinning body is the sum of all the pieces.

$$\therefore (K.E)_{rot} = \frac{1}{2} m_1 r_1^2 \omega^2 + \frac{1}{2} m_2 r_2^2 \omega^2 + \text{------} + \frac{1}{2} m_n r_n^2 \omega^2$$

$$= \frac{1}{2} \left(m_1 r_1^2 \omega^2 + m_2 r_2^2 \omega^2 + \text{------} + m_n r_n^2 \omega^2 \right)$$

$$= \frac{1}{2} \left(\sum_{i=1}^{n} m_i r_i^2 \right) \omega^2$$

$$\boxed{(K.E)_{rot} = \frac{1}{2} I \omega^2}$$

where $I = \sum_{i=1}^{n} m_i r_i^2$ = Moment of inertia of Body

Advantage Of Rotational K.E:

Rotational K.E has practical uses. For example, the rotational K.E of flywheels is used in many engines for keeping smooth rotation. A flywheel stores energy between the power strokes of the pistons, so that the energy is distributed over the full revolution of the crankshaft and hence rotation remains smooth.

Rotational K.E Of Disc And Hoop:

As, we know that rotational K.E of a spinning or rolling body is,

$$(K.E)_{rot} = \frac{1}{2} I \omega^2$$

The moment of inertia of a disc is given by,

$$I = \frac{1}{2} m r^2$$

$$\therefore (K.E)_{rot} = \frac{1}{2} \left(\tfrac{1}{2} m r^2 \right) \omega^2$$

$$= \frac{1}{4} m r^2 \omega^2$$

where, $V^2 = r^2 \omega^2$

Hence,

$$\boxed{(K.E)_{rot} = \frac{1}{4} m V^2}$$

This is expression for rotational K.E of a disc.

Also, the moment of inertia of a hoop is given by,

$$I = mk^2$$

$$\therefore (K.E)_{rot} = \frac{1}{2}(mk^2)\omega^2$$

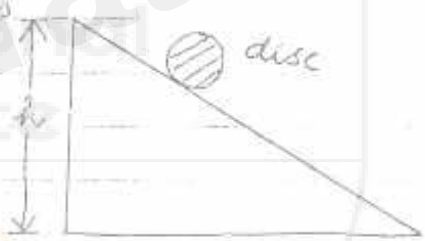
where, $v^2 = k^2\omega^2$

$$\boxed{(K.E)_{rot} = \frac{1}{2}mv^2}$$

This is expression for rotational K.E of a hoop.

Velocity Of A Disc And Hoop Rolling On An Incline Plane:

Consider a disk starts moving down on incline plane of height 'h'. The motion of disc consists of both rotational and translational motions. If no energy is lost due to friction, then total energy of the disc on reaching the bottom of incline plane must be equal its P.E at the top.



$\therefore$ Loss in P.E = Gain in Trans. K.E + Gain in rot. K.E

$$mgh = \frac{1}{2}mv^2 + \frac{1}{4}mv^2$$

Dividing B.S by 'm'

$$gh = \frac{1}{2}v^2 + \frac{1}{4}v^2$$

$$gh = \frac{3}{4}v^2$$

or $$v = \sqrt{4/3\,gh}$$

This is expression for velocity of disc on reaching bottom of incline plane.

FOR HOOP:

Loss in P.E = Gain in Trans. K.E + Gain in rot. K.E

$$mgh = \frac{1}{2}mV^2 + \frac{1}{2}mV^2$$

$$mgh = mV^2$$

$$V = \sqrt{gh}$$

This is expression for velocity of a hoop on reaching bottom of incline plane.

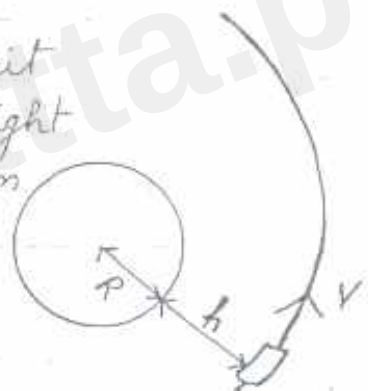
ARTIFICIAL SATELLITES:

Artificial satellites are the objects that revolve around the earth in certain orbits and the gravitational pull of earth provides the necessary centripetal force.

They are put into orbits by rockets.

EXPRESSION FOR ORBITAL VELOCITY:

Consider a satellite is revolving around the earth in an orbit of radius 'r' at very low height above the surface of earth. When a satellite is moving in a circular orbit, it has centripetal acceleration.



$$a_c = \frac{V^2}{r}$$

where, $r = R + h$ (R = radius of earth)

$r \approx R$

The centripetal acceleration is produced due to force of gravity

$$\therefore a_c = g$$

$$\boxed{V = \sqrt{gR}}$$

This is expression for the orbital velocity of the satellite.

where,

g = acc. due to gravity = $9.8 m/s^2$

R = radius of earth = 6.4×10^6 m

$$V = \sqrt{9.8 \times 6.4 \times 10^6}$$

$$= 7.9 \times 10^3 \text{ m/sec}$$

or $V = 7.9$ Km/sec

This is the minimum orbital velocity necessary to put a satellite into a stable orbit and is called centripetal velocity.

Time Period For Circular Motion Of A Satellite:

Time taken by satellite to complate one revolution around the earth is called "Time Period".

It is denoted by 'T' and given as,

$$T = \frac{\text{Distance}}{\text{Velocity}}$$

When a satellite revolve around earth at very low height, distance covered by it is,

$$\text{Distance} = 2\pi R$$

$$= \text{Circumference of earth}$$

$$T = \frac{2\pi R}{V}$$

$$= \frac{2(3.14)(6.4\times10^{6})}{7.9\times10^{3}}$$

$$T = 5060 \text{ sec} = 84 \text{ min approximately.}$$

When a satellite revolve around in an orbit at large distance 'h' above the surface of earth, then slower obital velocity is required and it will take long time to complete one revolution.

ADVANTAGE OF CLOSE ORBITING SAIELLITES:

Close orbiting satellites orbit the earth at a height of above 400 Km. Twenty four such satellites form the Global Positioning System.

An airline pilot, sailor or any person can used a pocket size instrument or mobile phone to find his position on the earth surface to within 10 m accuracy.

REAL AND APPARENT WEIGHT:

(i) REAL WEIGHT:

The real weight of an object is equal to the gravitational pull of earth on the object.

Mathematically,

$$W = mg$$

where,

m = Gravitational mass of an object
g = acc. due to gravity = 9.8 m/sec^2.

(iii) APPARENT WEIGHT:

The weight of an object in a certain frame of reference is equal and opposite to the force needed to prevent the object from falling from rest in that frame of reference.

Its value depends upon the acceleration of the frame of reference.

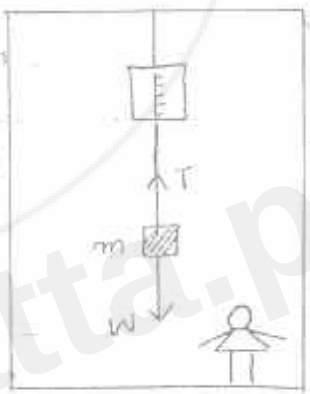
EXPLANATION:

In order to explain the idea of apparent weight, let us consider the following example.

EXAMPLE:

Consider a body of mass 'm' tied to a spring balance which is attached to the ceiling of a lift as shown in figure.

The reading of the spring balance shows the tension in the string and is called apparent weight of the body. Its value depends upon the acceleration of the lift.



LIFT AT REST OR MOVING WITH UNIFORM VELOCITY:

When lift is at rest or moving with uniform velocity. Then,

According to Newton's 2nd law,

acc of object = $a = 0$

Also, Net force on object = $F = 0$

If 'W' is gravitational force acting on object and tension in string is 'T'. Then,

$$T - W = ma$$

$$T - W = m(0)$$

$$T - W = 0$$

$$T = W = mg$$



Hence apparent weight is equal to real weight.

Lift Moving Upward With Acceleration:

When lift is moving upward with an acceleration 'a' then, according to Newton's 2nd law,

$$T - W = F$$

As,

$$F = ma \quad \text{and} \quad W = mg$$

$$\therefore T - mg = ma$$

$$T = ma + mg$$

$$T = m(g + a)$$



The apparent weight of an object is more than its real weight by an amount ma.

Lift Moving Downward With Acceleration:

Suppose the lift is moving downward with an acc. 'a' Then, according to Newton's 2nd law

$$W - T = F$$

As

$F = ma$ and $W = mg$

$\therefore mg - T = F$

$mg - T = ma$

$\Rightarrow T = mg - ma$

$T = m(g - a)$



So, apparent weight of an object is less than its real weight by an amount ma.

Lift Is Falling Freely Under Gravity:

Suppose the lift is falling freely under gravity with acceleration $a = g$ then the apparent weight of an object is given by

$T = W - F$

As,

$W = mg$ and $F = ma$

$\therefore T = mg - ma$

$T = m(g - a)$

But here,

$a = g$

$\therefore T = m(g - g)$

$T = m(0)$

$T = 0$

So, apparent weight becomes zero. Thus according to this observation, the body appears to weight less.

Weightlessness In Satellite And Gravity Free System:

When a satellite or

space-craft is falling freely in space, everything within this freely falling system will appear to weightless. Thus no force is required to prevent an object from falling in the frame of reference of the space-craft.

Such a system is called gravity free system.

REASON FOR WEIGHTLESSNESS:

When we stand on the ground we feel the upward contact force or reaction force of ground. This falling tells us that our feet are pressing against the ground. If the ground is taken away from under our feet so we are falling through air then we have no experience of any contact force on us and so we feel weightless. The astronaut who is in a space-craft orbiting the earth is in a permanent state of free fall towards the earth. This can be explain as follow:

EXPLANATION:

Consider a projectile is thrown parallel to earth surface at increasing speeds then curvature of path of projectile decreases during its free fall on earth. If the projectile (space-craft) is thrown with certain large velocity parallel to earth, the curvature of its path will match the curvature of earth. In this case, the space-ship will simple circle the earth.

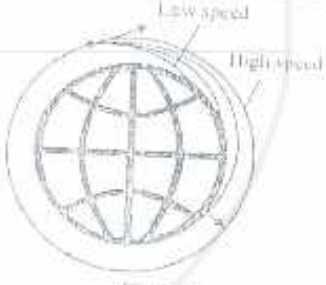


Fig 5.1

The astronauts and their space-ships is falling freely towards the earth.

with acceleration due to gravity ($g = 9.8\ m/sec^2$) all the time but the curvature of earth prevents the space-craft from hitting. So there is no contact force between the astronaut and space-ship. As, he feel no support or contact force from the space-ship around him, he experiences apparent weightlessness.

ORBITAL VELOCITY:

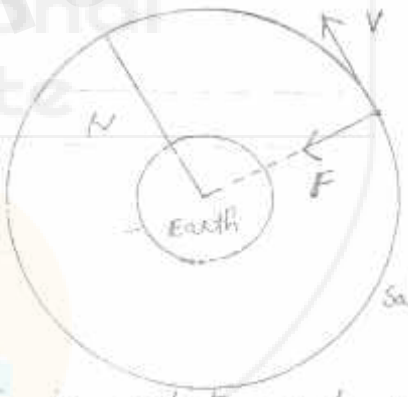
The artificial satellites revolve around the earth in nearly circular orbits. The instantaneous velocity of the satellite is along the tangent to circular orbit and is called orbital velocity.

EXPRESSION FOR ORBITAL VELOCITY:

Consider a satellite revolving around the earth in a circular orbit of radius r.

Let,

M = mass of earth
m_s = mass of satellite
V = Orbital velocity of satellite



The centripetal force is required to hold the satellite in orbit and given as,

$$F_c = \frac{m_s V^2}{r} \longrightarrow (i)$$

This force is provided by the gravitational force of attraction between the earth and the satellite.

$$\text{Gravitational Force} = \frac{G m_s M}{r^2} \longrightarrow (ii)$$

Comparing expression (i) and (ii)

$$\frac{m_s v^2}{r} = G\frac{m_s M}{r^2}$$

$$\Rightarrow \quad v = \sqrt{\frac{GM}{r}}$$

This is expression for orbital velocity of satellite. This shows that orbital velocity is independent of the mass of satellite.

Significance Of Orbital Velocity:

A satellite orbiting the earth in a circular orbit must have the orbital velocity given by,

$$v = \sqrt{\frac{GM}{r}}$$

where, r = distance of satellite from centre of earth

The orbital velocity is the minimum velocity necessary to hold a satellite into a stable orbit. It is also called "critical velocity". Any velocity less than critical value will bring the satellite tumbling back to the earth.

Artificial Gravity:

When a satellite moves on a circular orbit around the earth in deep space, the astronauts in it is in the state of weightlessness. If the satellite is a space laboratory then it is very dizzicult to the astronaut to stay long in that state and perform experiments. In order to overcome this dizzicult, artificial gravity is created. It is done by rotating

the satellite about its own axis with certain frequency.

DETERMINATION OF FREQUENCY:

Consider a space ship rotating about its axis with constant speed 'V'. The outer radius of space-ship is R.

Any person who is standing at the outer rim will experience centripetal acceleration

$$a_c = \frac{V^2}{R}$$

$$a_c = \frac{(R\omega)^2}{R} \qquad \because (V = R\omega)$$

$$\Rightarrow \quad a_c = R\omega^2$$

Also, time period of revolution of space-ship is,

$$T = \frac{2\pi}{\omega}$$

or $$\omega = \frac{2\pi}{T}$$

Hence,

$$a_c = R\left(\frac{2\pi}{T}\right)$$

$$a_c = R\,\frac{4\pi^2}{T^2}$$

As,

$$f = \frac{1}{T}$$

$$\therefore a_c = 4\pi^2 R f^2$$

$$f^2 = \frac{1}{4\pi^2}\,\frac{a_c}{R}$$

$$f = \frac{1}{2\pi}\sqrt{\frac{a_c}{R}}$$

As, we know that, when satellite moves in the orbit around the earth, the force of the necessary centripetal acceleration so that.

$$a_c = g$$

$$\therefore f = \frac{1}{2\pi}\sqrt{\frac{g}{R}}$$

When the space-ship rotates with this frequency, the artificial gravity like earth is provided to astronauts in space-ship.

Mechanism Of Artificial Gravity:

When we stood on the ground we feel the upward constant force or reaction force of the ground. This feeling feels us that our feet are pressing against the ground and there is a weight force acting on us.

When the space-ship is set into rotation about its own axis, the astronaut is pressed towards the outer rim and exerts a force on the floor of space-ship in same way as on the earth. This way artificial gravity like earth is created in space-ship.

Geostationary Orbits:

Satellites are used for international communicational links. Communicational satellites are often placed in a high orbit over the equator. The orbit is called a geosynchronous or geostationary orbit.

In this orbit, the orbital motion of

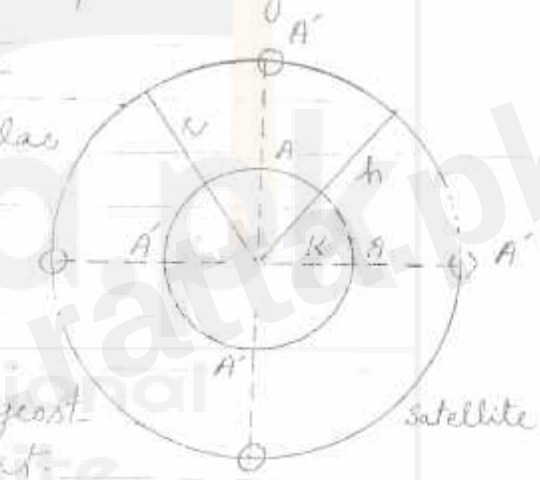
satellite is synchronized with the rotation of the earth. So the satellite remains always over the same point on the equator as the earth spins on its axis. The high orbit allows a large area of earth to be seen by the satellite.

EXPRESSION FOR RADIUS OF GEOSTATIONARY ORBIT:

Let us find the expression for radius of geostationary orbit.

As we know that the orbital speed for circular orbit is given by,

$$V = \sqrt{\frac{GM}{r}} \longrightarrow (i)$$



where r = radius of geostationary orbit.

The average speed of satellite in one day is given by,

$$V = \frac{S}{t}$$

$$V = \frac{2\pi r}{T} \longrightarrow (ii)$$

where T = Time period of motion of satellite = one day.

As, the earth rotates in one day and the satellite revolve around the earth in one day. The satellite at A' will always stay over the same point A on the earth as shown in

fig.

Comp ① and (ii), we get.

$$\frac{2\pi r}{T} = \sqrt{\frac{GM}{r}}$$

Squaring B.S, we get.

$$\frac{4\pi^2 r^2}{T^2} = \frac{GM}{r}$$

$$r^3 = \frac{GMT^2}{4\pi^2}$$

$$\boxed{r = \left(\frac{GMT^2}{4\pi^2}\right)^{1/3}}$$

This is expression for orbital radius of geostationary orbit.

where,

G = Universal gravitational constant = $6.67 \times 10^{-11}\ N \cdot m^2/kg^2$

M = Mass of earth = 6×10^{24} kg

T = Time period of orbital motion of satellite

= one day

= 24 hours

= 24 × 60 × 60 seconds

Substitute these value in above eq. we get.

$r = 4.23 \times 10^7$ m

$r = 4.23 \times 10^4$ Km

The height of geostationary orbit above the eq. is,

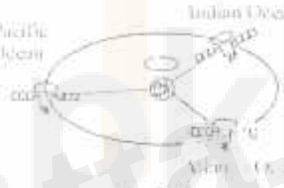
$h = r - R = 4.23 \times 10^4 - 6.4 \times 10^3 \approx 36000$ Km

COMMUNICATION SATELLITES:

Satellites are used for international communication links. A satellite communication system can be set up by placing

satellite in a high orbit over the equator at different points. This orbit is called geostationary orbit.

One such geostationary satellite covers 120 of longitude so that whole of the populated earth's surface can be covered by three correctly positioned satellites as shown in figure. Since the geostationary satellites above the same point on the earth, continuous communication with any place on the surface of earth is possible. Microwaves are set up to a satellite at one frequency say 4 GHz. and are then amplified and converted to another frequency say 11 GHz before being sent back down to a receiving station on the ground.



The range of frequencies available is between 3 and 30 GHz.

Microwaves are used in satellite communication due to following reasons:

(i) They travel in a narrow beam

(ii) They travel in a straight line.

(iii) They can easily pass through the atmosphere of earth. The energy needed to amplify and retransmit signals is provided by large solar cell panels fitted on the satellites.

There are over 200 earth stations which transmit signals to satellites and receive signals via satellites from other countries.

LARGEST SATELLITE SYSTEM:

The largest satellite system is managed by 126 countries, International Telecommunicational satellite organization. (INTELSAT)

An intelsat VI satellite appears at microwave frequencies of 4, 6, 11 and 14 GHz and has a capacity of 30,000 two way telephone circuit plus three TV channels.

Newton's And Einstein's Views Of Gravitation:

According to Newton, every two particles of matter attract each other with a force called force of gravitation.

The gravitation is the intrinsic property of the matter and is given by Newton's "law of gravitation".

According to this Law, "Every particle of matter attracts every other particle with a force which is directly proportional to the product of their masses and inversly proportional to the square of the distance between their centres".

Einstein's Views Of Gravitation:

According to Einstein's theory "space time is curved near massive bodies".

Explanation:

In order to explain Einstein's idea, let us consider space as a thin rubber sheet. If a heavy weight is hung from thin rubber sheet, it curves as shown in figure.

The weight corresponds to a huge mass that causes space itself to curve. In Einstein's theory we do not speak of force of gravity acting on bodies, rather we say that, "Material particles and light rays move along geodesics in curved space time". So a body at rest or moving slowly

near the great mass would follow a geodesic towards that body.

Comparison Of Eistein's And Newton's Theory Of Gravitation:

Einstein's theory gives us a physical picture of how gravity works.

Newton discovered the inverse square law of gravity but did not offer an explanation of why gravity follows an inverse square law.

Einstein's theory says that gravity follows an inverse square law (except in strong gravitational fields). It also tells us why gravity follows an inverse square law. Therefore, we can conclude that Einstein's theory is better than Newton's even though it includes Newton's theory within itself and gives the same answers as Newton's theory every-where except where the gravitational field is very strong.

Success Of Einstein's Theory Of Gravitation:

According to Einstein "If gravity and acceleration are precisely equivalent, gravity must bend light by a precise amount which can be calculated".

This was not entirely new idea, the same idea was given by Newton in his Corpuscular theory of light. According to Newton's corpuscular theory of light Light consists of a stream of tiny particles called corpuscles. These tiny particles would be deflected by gravity.

But in Einstein's theory, the deflection of light by gravity is predicted to be twice than that predicted by Newton's theory.

The bending of star light by gravity of sun was measured during solar eclipse in 1919. and found to match Einsteins prediction rather than Newton's. Light from the star A is deflected as it passes close to the sun on its way to direction B, shifted by the angle ϕ. Einstein predicted that $\phi = 1.74''$ seconds of angle which was found to be same during the solar eclipse of 1919.

